

Natural Language Processing (Elective)

Course Objectives:

- To provide a well grounded foundation in theories, models and algorithms of natural language processing (NLP)
- To show how NLP systems deal with representing and processing words, syntax and semantics
- To develop an understanding of the applications and research areas of NLP

Contents

- | | |
|--|-----------------|
| 1. Introduction | [3 hrs] |
| 1.1. Language, thought and understanding | |
| 1.2. Ambiguity | |
| 1.3. History | |
| 1.4. State of art and future | |
| 2. Morphology | [7 hrs] |
| 2.1. Regular Expressions and Automata | |
| 2.2. Morphology and Finite-State Transducers (FST) | |
| 2.3. The Lexicon and Morphotactics | |
| 2.4. Lexicon-Free FSTs: Porter Stemming | |
| 2.5. N-grams | |
| 3. Part-of-Speech (PoS) Tagging | [10 hrs] |
| 3.1. Rule-based PoS Tagging | |
| 3.2. Stochastic PoS Tagging | |
| 3.3. PoS Tagging and Hidden Markov Models (HMM) | |
| 4. Syntactic Analysis | [12 hrs] |
| 4.1. Constituency (Phrase-Level , Sentence-Level) | |
| 4.2. Context-Free Grammars (for English) | |
| 4.3. Top-down and bottom-up parsing | |
| 4.4. Features and Unification | |
| 4.5. Lexicalised and Probabilistic Parsing | |
| 4.6. Language and Complexity – the Chomsky hierarchy | |
| 5. Semantics | [12 hrs] |
| 5.1. Representing Meaning | |
| 5.2. Meaning structure of language | |
| 5.3. First Order Predicate Calculus and semantics | |
| 5.4. Syntax-driven Semantic analysis | |
| 5.5. Lexical Semantics | |
| 5.6. WordNet: A database of lexical relations | |
| 5.7. Word Sense Disambiguation | |
| 6. Pragmatics | [10 hrs] |
| 6.1. Reference resolution | |
| 6.2. Text coherence | |
| 6.3. Discourse structure | |

7. Applications

[6 hrs]

- 7.1. Information Extraction
- 7.2. Question Answering and Summarization
- 7.3. Dialog and Conversational Agents
- 7.4. Sentiment Analysis
- 7.5. Machine Translation

References

- 1. Jurafsky, Daniel & James H. Martin. (2008) *Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition* (2nd edition). Pearson Education
- 2. Manning, Christopher D. & Hinrich Schütze. (1999) *Foundations of statistical natural language processing*. Cambridge: The MIT press.
- 3. Allen, J. (1995) *Natural Language Understanding* (2nd ed.). Benjamin/Cummins publishing.